

ForceField: Visualizing Intermaterial Interaction through Floor and Depth Sensing

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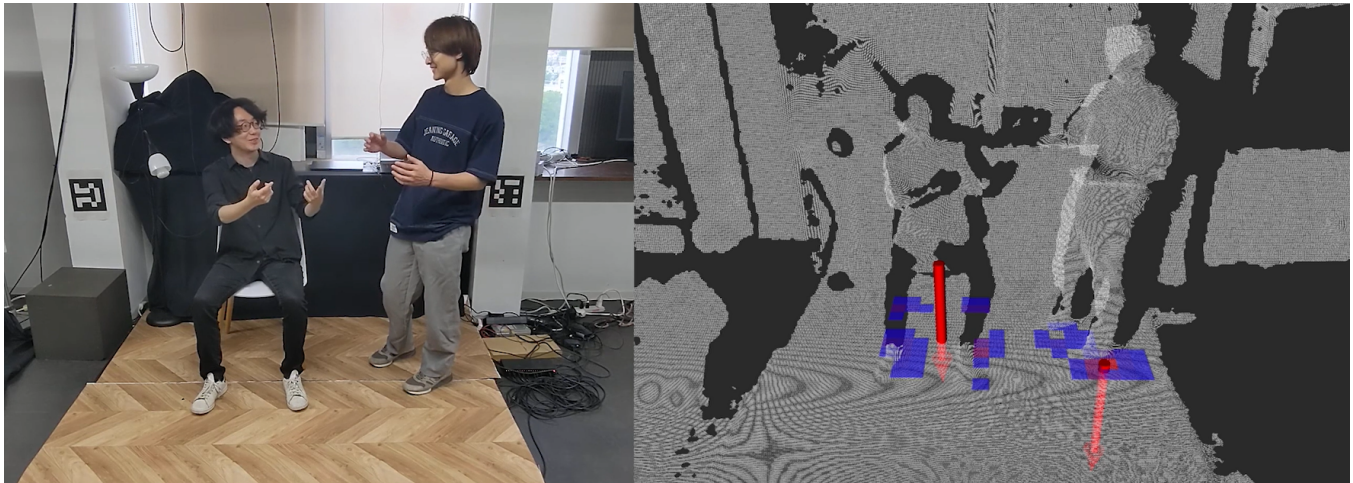


Figure 1: Overview of ForceField: (Left) one user is interacting with a chair object in living space, while another is standing straight on the floor (Right) The intermaterial interaction, the contact forces between the body and the object or the space, are captured and visualized as the red arrows.

ABSTRACT

ForceField realizes the measurement of intermaterial contact forces across any surface within a room-scale environment without the need for additional device attachments to bodies or objects. It achieves this by acquiring a 3D model through a depth sensor and capturing ground pressure through a floor sensor, and then reconstructing contact forces between surfaces where no sensors are attached, such as when a hand pushes a desk, or people push against each other, from information on force and geometry. Through this unobtrusive environmental measurement approach, ForceField opens up the possibility of highly physical spatial interactions in domains such as healthcare, sports, and entertainment.

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1 INTRODUCTION

Humanity has always sought to see the unseen by visualizing the invisible. This is also true in human-computer interaction (HCI) research, where understanding invisible dynamics of human behaviors and interactions with objects plays a crucial role [Ishii and Ullmer 1997; Sato et al. 2004]. The forces present within a space serve as a vital source of information for capturing events occurring within that space in detail. We expect a significant improvement in our understanding of behavior within a space by measuring and visualizing information on the forces at the boundaries between the materials, as shown in Fig. 1.

Despite the potential usefulness of force visualization, there are still many challenges to promoting their acquisition and utilization. This study aims to provide spatial force information in a form as accessible and adaptable as geometric information. To achieve this goal, we propose a visualization framework for spatial contact forces and explore the potential for interaction applications in space.

In this study, we propose a new framework, ForceField, for understanding and visualizing contact forces within a space. ForceField

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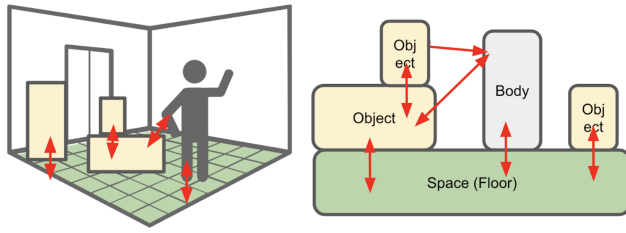


Figure 2: The concept of ForceField: The intermaterial interaction between body, object and space can be acquired without any additional devices attached on bodies or objects.

can measure and estimate contact forces at any boundary between bodies, objects, and spaces within a 3D space in real-time by integrating measurement data of force and geometry using sensor fusion technology. This data of force measurement in 3D space has various application possibilities, not limited to HCI.

2 FORCEFIELD CONCEPT

The concept of ForceField is the acquisition and visualization of intermaterial interaction within a space through the integration of two major physics modalities, force and geometry. Here, intermaterial refers to the contact forces acting between the mutual relationships of body, object, and space, as shown in Fig 2. The technological advantage of ForceField is to remove inserting pressure sensors into contact point between bodies and objects, which has significantly impaired implementation.

For objects and human bodies in direct contact with environmental sensors, the sensor output information can be used as intermaterial interaction as it is. For objects and human bodies not in direct contact with environmental sensors, we estimate intermaterial interactions such as contact state and contact force by utilizing geometric information and mechanical conservation laws [Yoshida et al. 2019]. For exploring more application areas, visualization can be done not only through a 2D flat display but also through VR/AR glasses, promising a more immersive visualization experience.

3 PROTOTYPE IMPLEMENTATION

The experimental space was set at 2.0 x 1.5m, and simultaneous acquisition of geometry and pressure was performed using a depth sensor and a ground pressure sensor. As environmental sensors, we used Azure Kinect [Kean et al. 2011] for geometric signal acquisition and Flexel [Yoshida et al. 2022] ground pressure sensor for tactile signal acquisition.

We measured the pressure distribution, and calculated total force within focused spatial segments. We identified the force position in the space by finding the intersection between the line of action and the environmental geometry. In our prototype, we have constructed precomputed coarse 3d model from geometric sensing, we mapped the contact point onto the the environmental mesh.

4 POTENTIAL APPLICATION

As ForceField offers an environmental measurement that does not require body wearables or furniture attachments, it has various applications, as shown in Fig 3.

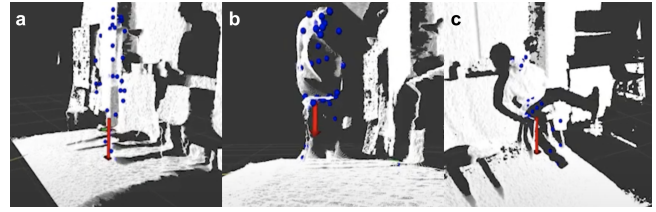


Figure 3: Example visualized sequence in intermaterial interactions: (a) body-space interaction in walking, and (b-c) body-object interaction during pushing and seating.

Body-Body. ForceField aids in skill transmission like rehabilitation by visualizing therapist-applied forces for enhanced learning.

Body-Object. The technology turns any object surface into a touch sensor, enabling versatile anywhere surface interaction.

Body-Space. Useful in healthcare and sports, ForceField monitors body-space interactions, such as fall detection and gait recognition.

Object-Object. The system tracks interactions between objects, alerting users about misplaced items like a forgotten smartphone.

Object-Space. ForceField aids in space management by recognizing furniture layouts based on ground-force data.

5 CONCLUSION

We proposed ForceField, a novel visualization method of intermaterial interaction through force and geometry sensing. We also plan to continue exploring through evaluation experiments and regarding application areas.

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REFERENCES

- Hiroshi Ishii and Brygg Ullmer. 1997. Tangible bits: towards seamless interfaces between people, bits and atoms. In *Proceedings of the ACM SIGCHI Conference on Human factors in computing systems*. 234–241.
- Sean Kean, Jonathan C Hall, Phoenix Perry, Sean Kean, Jonathan C Hall, and Phoenix Perry. 2011. Microsoft's kinect sdk. *Meet the Kinect: An Introduction to Programming Natural User Interfaces* (2011), 151–173.
- Tomomasa Sato, Tatsuya Harada, and Takatoshi Mori. 2004. Environment-type robot system "RoboticRoom" featured by behavior media, behavior contents, and behavior adaptation. *IEEE/ASME Transactions on Mechatronics* 9, 3 (2004), 529–534.
- Takatoshi Yoshida, Narin Okazaki, Ken Takaki, Masaharu Hirose, Shingo Kitagawa, and Masahiko Inami. 2022. Flexel: A Modular Floor Interface for Room-Scale Tactile Sensing. In *Proceedings of the 35th Annual ACM Symposium on User Interface Software and Technology*. 1–12.
- Takatoshi Yoshida, Xiaoyan Shen, Koichi Yoshino, Ken Nakagaki, and Hiroshi Ishii. 2019. SCALE: Enhancing Force-based Interaction by Processing Load Data from Load Sensitive Modules. In *Proceedings of the 32nd Annual ACM Symposium on User Interface Software and Technology*. 901–911.